

Midterm Practice Problems

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Student Name/Number: (doubleclick to edit)

Instructions

The following are example directions for an exam.

- Ensure you modify the field above with your **name and student number above immediately**
- The exam has **XXXXXX questions**, each with multiple parts for a total of **XXXXXX points**. You may not finish the exam, so best to do your best answering all questions to the extent possible and not get stuck on any one question.
- This exam is **closed book** and accessing the internet **is not permitted**
 - See the **formula “sheet”** embedded at the end of this notebook for reference
 - You can the internal help as required (see the Jupyterhub menu, use **Settings/Show Contextual Help**)
- **Execute the file** to begin, which will also check your setup. To do this in Jupyter, in the menu go > **Run > Run All Cells** or the equivalent buttons
- Edit this file directly, and in-place as an **ipynb** file, which we will automatically download at the end of the exam time directly. In particular
 - **DO NOT** rename this file with your name. It is automatically associated with your canvas account
 - **DO NOT** save-as the file, move it, or export to pdf or html
 - **DO NOT** add any additional packages
- **Save** the notebook as you are working
 - We will only grade what is saved at the end of the exam in this exact file, and it is your responsibility to ensure the file is saved
 - We will **not** execute the notebook, so ensure all code, figures, etc. are ready as-is upon saving for submission

- Ensure you **edit the results in the code blocks or markup blocks indicated** as we will not grade anything outside of those
 - You will not be judged on code quality directly, but code clarity may be required for us to ensure you understood the problem directly
 - If a question requires math, you can try to put latex inside of the cells but will not be judged on whether you write latex vs. math in text that doesn't quite match latex. But it should be clear

```
# Packages available
# DO NOT MODIFY OR ADD PACKAGES
using Distributions, Plots, LaTeXStrings, LinearAlgebra, Statistics, Random
```

Take the followign functions from our previous lectures

```
# DO NOT MODIFY
function lorenz(v) # assumed sorted vector
    S = cumsum(v) # cumulative sums: [v[1], v[1] + v[2], ... ]
    F = (1:length(v)) / length(v)
    L = S ./ S[end]
    return (; F, L) # returns named tuple
end
# Assumes that v is sorted!
gini(v) = (2 * sum(i * y for (i,y) in enumerate(v))/sum(v)
          - (length(v) + 1))/length(v)

function simple_regression(x, y)
    x_bar = mean(x)
    y_bar = mean(y)
    a = sum((x .- x_bar).*(y .- y_bar)) / sum((x .- x_bar).^2)
    b = y_bar - a * x_bar
    return (;a, b)
end
```

simple_regression (generic function with 1 method)

Short Question 1

With a CDF of a distribution $F(x)$ with an accompanying counter-cdf $1 - F(x)$, what is the definition of a power-law tail? For a stochastic process X_t , what features of that stochastic process might lead to a power-law tail in the stationary distribution?

(double click to edit your answer)

Short Question 2

If you are trying to understand wealth inequality, what features of the distribution are ideal to examine using (a) Lorenz Curves; (b) Gini Coefficients; and (c) Power Law Tails

(double click to edit your answer)

Short Question 3

If I calculate a log-log plot of a distribution to examine whether it has heavy tails, what should I plot for the x-axis and y-axis? If I did a linear regression and found that the slope was -2, would that necessarily say that the distribution was a power-law distribution? Why or why not?

(double click to edit your answer)

Short Question 4

If a log-log plot showed a slope of -1.5 with a clear linear relationship in the right tail, what is the implied power-law exponent? What does this tell us about moments of the underlying distribution?

(double click to edit your answer)

Short Question 5

Compare stochastic processes:

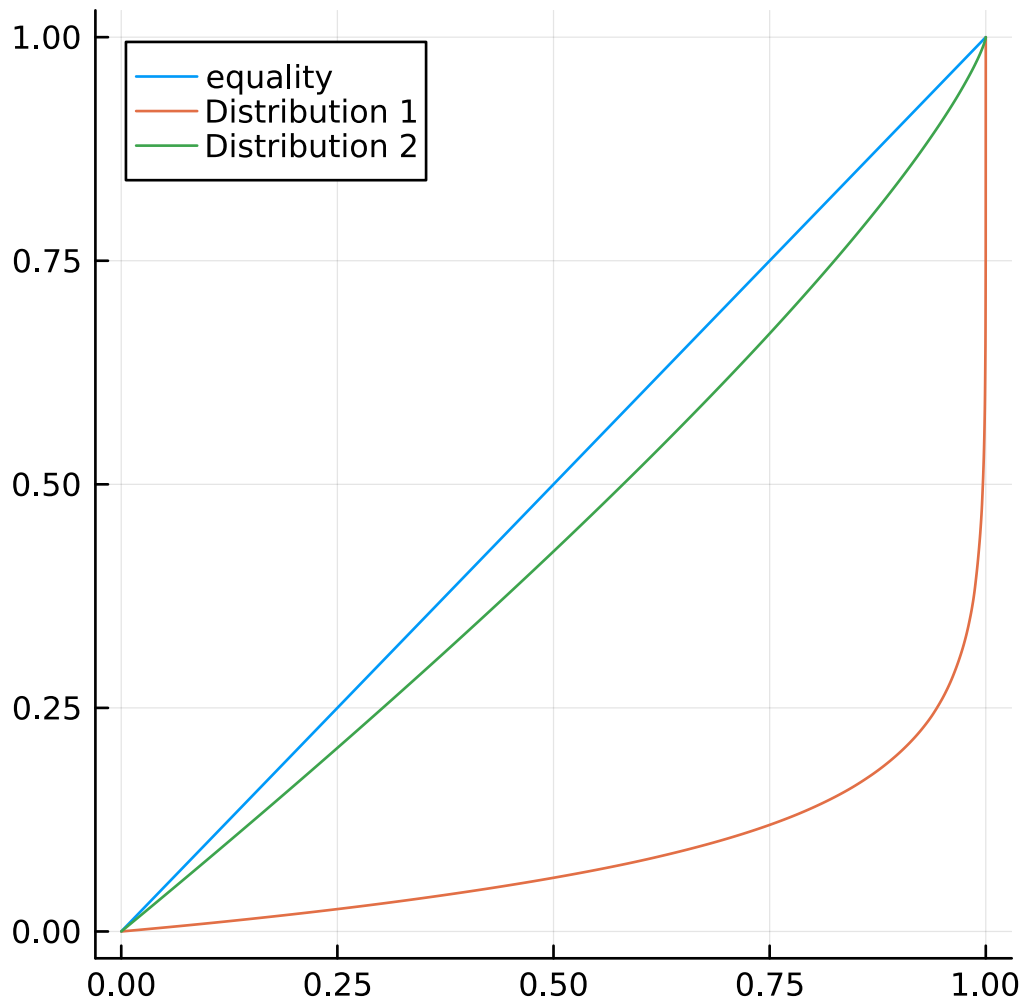
1. $X_{t+1} = a_{t+1}X_t + b$
2. $X_{t+1} = X_t + a_{t+1} + b$

where in both cases a_{t+1} are IID shocks. Which stochastic process would you expect may have power-law tails in their stationary distribution, and why? What if $b = 0$ in both cases? Provide intuition for the differences.

(double click to edit your answer)

Short Question 6

Take the following figure showing the Lorenz Curves for data from two different distributions. What can you infer about the inequality in the two distributions? What would you expect the Gini coefficients to be for the two distributions?



(double click to edit your answer)

Short Question 7

Given a mapping $F : \mathbb{R}^N \rightarrow \mathbb{R}^N$ what is the definition of a fixed point x^* of this mapping? Does it always exist?

(double click to edit your answer)

Short Question 8

Explain the importance of the fixed factor of production, land, in the Malthusian growth model

(double click to edit your answer)

Short Question 9

An asset pays off a deterministic stream of income, $\{y_{t+j}\}_{j=0}^{\infty}$, and a risk-neutral agent prices this using a discount factor $0 < \beta < 1$. The PDV of the income is then $p_t = \sum_{j=0}^{\infty} \beta^j y_{t+j}$

The agent prices this asset using the recursive formulation

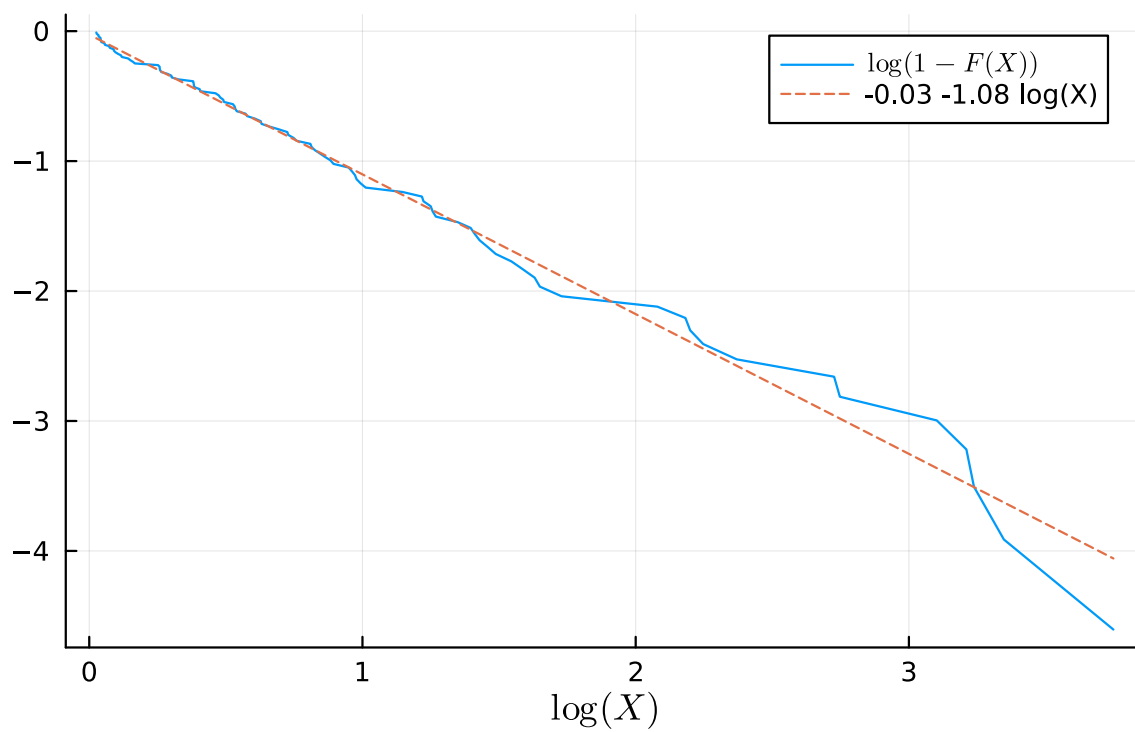
$$p_t = y_t + \beta p_{t+1}$$

Briefly interpret this recursive equation from the agent's perspective.

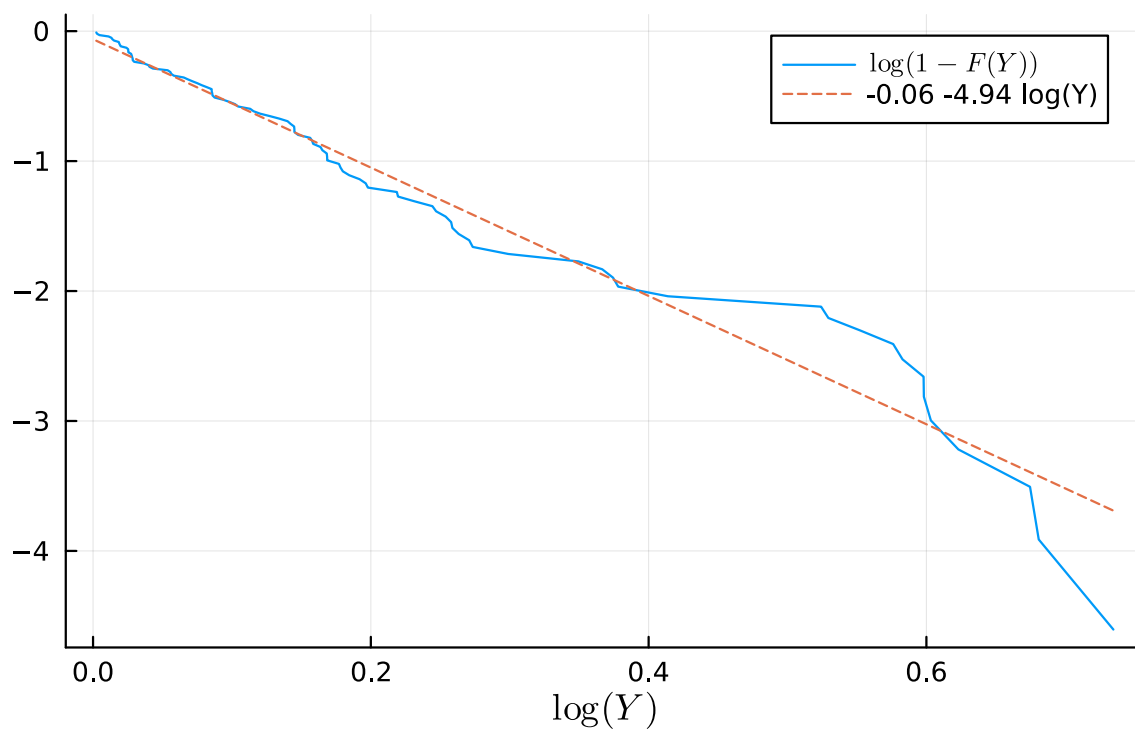
(double click to edit your answer)

Question 0

Consider for some $X_t \sim F$ IID draws from a distribution the log-log plot using the empirical CDF. The dotted line plots the linear regression with this data, with the intercept and slope shown on the legend.



Consider the log-log plot of 100 IID draws from a different distribution $Y_t \sim F$.

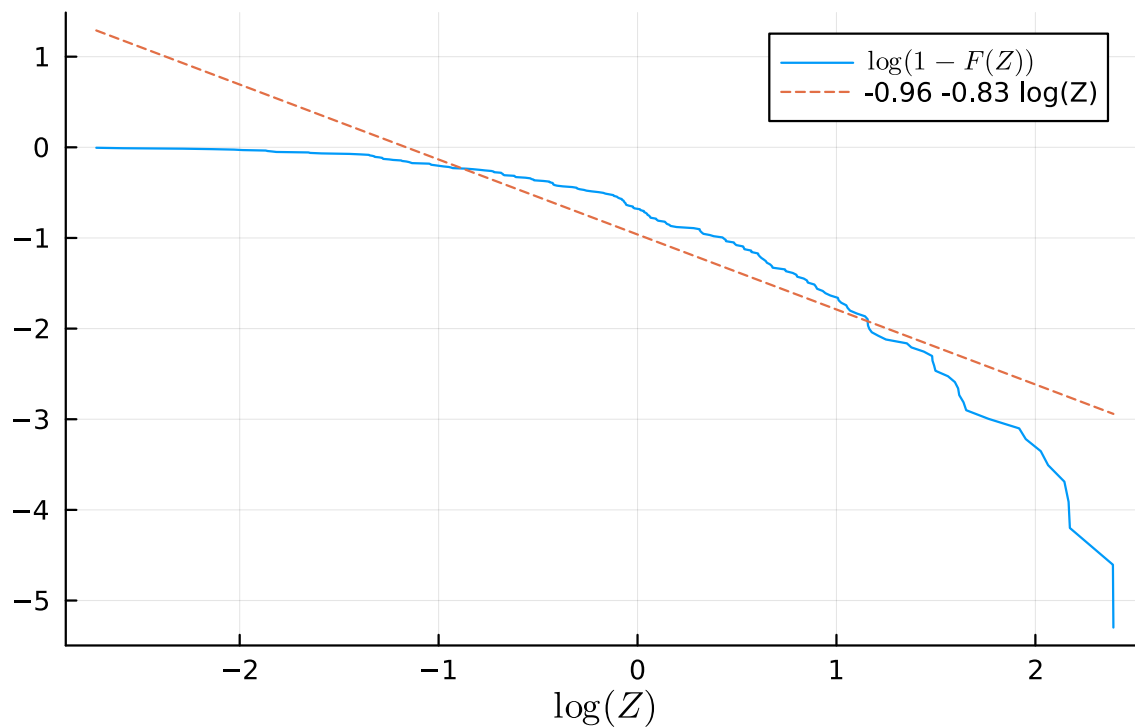


Compare the X_t and Y_t log-log plots. What can you infer about these distributions?

(double click to edit your answer)

Now consider the log-log plot of draws from a third distribution $Z_t \sim F$

The log-log plot of the Z_t draws



What can you infer about this distribution compared to the other two?

(double click to edit your answer)

Question 1

Following the notes on [AR\(1\) processes](#) rather than plotting the distribution as normal instead lets see what the stationary distribution looks like with simulation.

Part (a)

From $X_0 = 1.0$ simulate up to $T = 2000$ using the process

$$X_t = aX_{t-1} + b + cW_t$$

Where $a = 0.98, b = 0.1, c = 0.02$.


```
T = 2000
X_0 = 1.0
a = 0.98
b = 0.1
c = 0.02
# Add code here
```

0.02

Part (b)

On the same graph plot the histogram of those simulated values (i.e., $\{X_0, \dots, X_T\}$) then plot the density of the stationary distribution calculated in closed form in [those notes](#) (i.e. create a normal distribution with $\mu^* = b/(1 - a)$ and $v^* = c^2/(1 - a^2)$)

Hint: `histogram(X, normed=true)` normalizes the empirical draws so they are a proper PMF.

```
# Add code here
```

Part (c)

Now, do the same plot using the 10th to 500th observations (i.e., $\{X_{10}, \dots, X_{500}\}$)

```
# Add code here
```

And then a separate plot using the rest

```
# Add code here
```

Compare how these line? Explain why each is better or worse?

(double click to edit your answer)

Part (d)

Change the `c` parameter to be 0.2.

Now repeat the display of the two histograms (1) split of the 10th to 500th ; and (2) the remainder

```

T = 2000
X_0 = 1.0
a = 0.98
b = 0.1
c = 0.2
# Add code here, creating new cells as required

```

0.2

Compare this to part (d)? Has the results changed and if so, then why?

(double click to edit your answer)

Question 2

In this problem we will examine wealth dynamics for a simpler setup. Log income, $\log y_t$ follows an AR(1) process,

$$\log y_{t+1} = \mu_y + \rho_y \log y_t + \sigma_y \epsilon_{1,t+1}$$

and

$$\log R_{t+1} = \mu_r + \sigma_r \epsilon_{2,t+1}$$

where $\epsilon_{1,t+1} \sim N(0, 1)$ and $\epsilon_{2,t+1} \sim N(0, 1)$ are IID shocks.

As in the wealth dynamics problem from before, the evolution of wealth is given by

$$w_{t+1} = R_{t+1} s(w_t) + y_{t+1}$$

where $s(w_t)$ is the exogenously given savings function from before,

$$s(w) = s_0 w \cdot \mathbb{1}\{w \geq \hat{w}\} = \begin{cases} s_0 w & w \geq \hat{w} \\ 0 & w < \hat{w} \end{cases}$$

For a constant $s_0 \in (0, 1)$ which is savings rate and $\hat{w} \geq 0$ is a minimum wealth threshold below which they do not save and instead consumer all of their income (since $w_{t+1} = y_{t+1}$ in that case).

Take the following simpler structure for holding the parameters of the model

```

function simple_wealth_dynamics_model(;
    w_hat=1.0, # savings parameter
    s_0=0.75, # savings parameter
    mu_y=0.1, # labor income parameter
    sigma_y=0.1, # labor income parameter
    rho_y=0.9, # labor income parameter
    mu_r=0.0, # rate of return parameter
    sigma_r=0.2, # rate of return parameter
)
    return (;w_hat, s_0, mu_y, sigma_y, rho_y, mu_r, sigma_r)
end

```

simple_wealth_dynamics_model (generic function with 1 method)

and a modified version of the simulate_panel function

```

function simulate_panel(N, T, p; y_0 = p.mu_y/(1-p.rho_y), w_0 = p.mu_y/(1-p.rho_y))
    # Setup initial conditions
    w = w_0 * ones(N) # start at same w_0
    logy = log.(y_0 * ones(N)) # start at same y_0
    logR = zeros(N) # not used in this exact example, but could be generalized

    # Preallocate next period states and R intermediates
    w_p = similar(w)
    logR_p = similar(w)
    logy_p = similar(w)

    # Temporary used in calculations
    savings_proportion = similar(w) # include constant and R_{t+1}

    for t in 1:T
        R_shock = randn(N)
        y_shock = randn(N)
        @inbounds for i in 1:N
            logy_p[i] = p.mu_y + p.rho_y*logy[i] + p.sigma_y*y_shock[i]
            logR_p[i] = p.mu_r + p.sigma_r*R_shock[i] # no autocorrelation but could referen
            savings_proportion[i] = (w[i] >= p.w_hat) ? p.s_0 * exp(logR_p[i]) : 0.0
            w_p[i] = savings_proportion[i]*w[i] + exp(logy_p[i])
        end
        # Step forward
    end
end

```

```

        w .= w_p
        logy .= logy_p
        logR .= logR_p
    end
    sort!(w) # sorts the wealth so we can calculate gini/lorenz
    F, L = lorenz(w)
    return (;w, y = exp.(logy_p), F, L, gini = gini(w))
end

```

`simulate_panel` (generic function with 1 method)

The following code shows a basic simulation,

```

p = simple_wealth_dynamics_model() # Or simple_wealth_dynamics_model(;w_hat = 2.0) to swap out
N = 10_000
T = 200
w_0 = 10.0

res = simulate_panel(N, T, p; w_0) # uses the default y_0 but overrides the w_0 default
@show res.gini, mean(res.w), mean(res.y); # show some of the results

```

```

(res.gini, mean(res.w), mean(res.y)) = (0.17299150875033556, 11.89817693904152, 2.79571179214

```

Part (a)

Using the above code and default parameter values, simulate the model to see the difference in the gini coefficients where you change the variance on the returns `sigma_r` 10 points between 0.0 to 0.3.

Using your simulations, plot the gini coefficient as a function of `sigma_r` and comment on the results.

```

# Modify/add code here, creating new cells as required
sigma_r_values = range(0.0, 0.3, 10)
N = 10_000
T = 200

# simple_wealth_dynamics_model(;sigma_r = sigma_r_values[1]) # etc. to create modified model.
res = simulate_panel(N, T, simple_wealth_dynamics_model(;sigma_r = sigma_r_values[end])) # f
@show res.gini;

```

```
res.gini = 0.23481426677063247
```

Can you provide a brief interpretation of the results?

(double click to edit your answer)

Part (b)

Now, take the same model and let's shut off all variation on the income process to leave it as a fixed value so $y_{t+1} = y_t$ by setting `sigma_y = 0`, `rho_y = 1`, `mu_y = 0` and then initializing all of the agents with the `y_0 = 0.5` and `w_0 = 2.0`

Calculate the gini coefficient for the same range of `sigma_r` values as above and plot the results.

```
# Modify/add code here, creating new cells as required
p = simple_wealth_dynamics_model(;sigma_y = 0.0, rho_y = 1.0, mu_y = 0.0)
N = 10_000
T = 1000
y_0 = 0.5
w_0 = 2.0
# res = simulate_panel(N, T, p; y_0, w_0) # adapt this, passing in the y_0 and w_0
```

2.0

Can you provide a brief interpretation of the results?

(double click to edit your answer)

Formulas

Use the following formulas as needed. Formulas are intentionally provided without complete definitions of each variable or conditions on convergence, which you should study using your notes.

YOU WILL NOT BE EXPECTED TO USE THE MAJORITY OF THESE FORMULAS - BUT SOME MAY HELP PROVIDE INTUITION

General and Stochastic Process Formulas

Description 1	Formula 1	Description 2	Formula 2
Partial Geometric Series	$\sum_{t=0}^T c^t = \frac{1-c^{T+1}}{1-c}$	Geometric Series	$\sum_{t=0}^{\infty} c^t = \frac{1}{1-c}$
PDV	$p_t = \sum_{j=0}^{\infty} \beta^j y_{t+j}$	Recursive Formulation of PDV	$p_t = y_t + \beta p_{t+1}$
Univariate Linear Difference Equation	$x_{t+1} = ax_t + b$	Solution	$x_t = b \frac{1-a^t}{1-a} + a^t x_0$
Linearity of Normals	$X \sim \mathcal{N}(\mu_X, \sigma_X^2), Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ then $aX + bY \sim \mathcal{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2)$	Special Case	$Y \sim N(\mu, \sigma^2)$ then $Y = \mu + \sigma X$ for $X \sim N(0, 1)$
Partial Sums $X_1, \dots$ IID with $\mu \equiv \mathbb{E}(X)$	$\bar{X}_n \equiv \frac{1}{n} \sum_{i=1}^n X_i$	Strong LLN	$\mathbb{P}(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$
AR(1) Process	$X_{t+1} = aX_t + b + cW_{t+1}$ with $W_{t+1} \sim \mathcal{N}(0, 1)$	Stationary Distribution	$X_{\infty} \sim \mathcal{N}(\frac{b}{1-a}, \frac{\sigma_{\epsilon}^2}{1-a^2})$
AR(1) Evolution	$X_t \sim \mathcal{N}(\mu_t, v_t)$, then $X_{t+1} \sim \mathcal{N}(a\mu_t + b, a^2v_t + c^2)$	Recursively	$\mu_{t+1} = a\mu_t + b, v_{t+1} = a^2v_t + c^2$
Mean Ergodic Definition Given Stationary X_{∞}	$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T X_t = \mathbb{E}[X_{\infty}]$	ARCH (1)	$X_{t+1} = aX_t + (\beta + \gamma X_t^2)^{1/2} W_{t+1}$ with $W_{t+1} \sim \mathcal{N}(0, 1)$

Inequality and Power Law Formulas

Description 1	Formula 1	Description 2	Formula 2
Kesten Process for a_{t+1}, y_{t+1} IID	$X_{t+1} = a_{t+1}X_t + y_{t+1}$	Key Conditions for Kesten Stationarity	$\mathbb{E}(\log a_t) < 0$ and $\mathbb{E}(y) < \infty$
Counter-CDF	$\mathbb{P}(X > x) = 1 - \mathbb{P}(X \leq x)$	CCDF With density $f(x)$ and CDF $F(x)$	$\int_x^{\infty} f(x)dx = 1 - F(x)$
Pareto PDF with x_m minimum and tail parameter α	$f(x) = \frac{\alpha x_m^{\alpha}}{x^{\alpha+1}}$, for all $x \geq x_m$	CDF and CCDF	$F(x) = 1 - \left(\frac{x}{x_m}\right)^{-\alpha}$, $1 - F(x) = \left(\frac{x}{x_m}\right)^{-\alpha}$
Log-Log Plot with CDF $F(x)$	$\log(x)$ vs. $\log(1 - F(x))$	For Pareto	$\log(1 - F(x)) = \alpha \log(x_m) - \alpha \log(x)$

Description 1	Formula 1	Description 2	Formula 2
Power-law Tail	$\mathbb{P}(X > x) \propto x^{-\alpha}$ for large x	With CCDF	$1 - F(x) \propto x^{-\alpha}$ for large x
Empirical CDF	$\hat{F}(x) = \frac{\text{number of observations } X_n \leq x}{N}$	Tail Parameter Regression where $\alpha \approx -a$	$\log(1 - \hat{F}(x_i)) = b + a \log(x_i) + \epsilon_i$
Quantile Function	$x = F^{-1}(p) \equiv Q(p)$	Lorenz Curve	$L(p) = \frac{\int_0^p Q(s) ds}{\int_0^1 Q(s) ds}$
CDF with Ordered Data $v_1, \dots, v_n$	$F(v_i) \equiv F_i = \frac{i}{n}$	Lorenz with Ordered Data	$S_i = \frac{1}{n} \sum_{j=1}^i v_j$, $L(v_i) \equiv L_i = \frac{S_i}{S_n}$
Gini Coefficient	Area between Lorenz Curve and Line of Equality	Gini with Ordered Data	$G = \frac{2 \sum_{i=1}^n i v_i}{n \sum_{i=1}^n v_i} - \frac{n+1}{n}$

Solow and Stochastic Growth Formulas

Description 1	Formula 1	Description 2	Formula 2
Production	$Y_t = z_t F(K_t, N_t)$	Constant Returns to Scale	$F(\alpha K, \alpha N) = \alpha F(K, N) \quad \forall \alpha > 0$
Production per Capita	$k_t \equiv K_t/N_t, f(k_t) \equiv F(k_t, 1)$	Scaled Production Function	$f(k_t) = z_t f(k_{t-1})$
Marginal Product of Capital	$z_t \frac{\partial F(K_t, N_t)}{\partial K_t}$	MPK with $f(k) = k^\alpha$	$z_t \frac{\partial F(K_t, N_t)}{\partial K_t} = \alpha z_t k_t^{\alpha-1}$
Consumption/Investment	$C_t + X_t = Y_t \equiv z_t F(K_t, N_t)$	Capital Accumulation	$K_{t+1} = (1 - \delta)K_t + X_t$
Constant Population Growth	$N_{t+1} = (1 + g_N)N_t$	Per-Capital Evolution	$k_{t+1} = \frac{1}{1+g_N} [(1 - \delta)k_t + s z_t f(k_t)]$
Steady State	$(g_N + \delta)\bar{k} = s \bar{z} f(\bar{k})$	Steady State with $f(k) = k^\alpha$	$\bar{k} = \left(\frac{s \bar{z}}{g_N + \delta} \right)^{\frac{1}{1-\alpha}}$
Real Rental Rate of Capital	$r_t = z_t f'(k_t)$	Real Wages	$w_t = (1 - \alpha) z_t f(k_t)$
Stochastic Growth Capital Evolution	$k_{t+1} = (1 - \delta)k_t + s Z_t f(k_t)$, given k_0	Stochastic Growth Productivity Process	$\log Z_{t+1} = a \log Z_t + b + c W_{t+1}$ with $W_{t+1} \sim \mathcal{N}(0, 1)$